Abhishek Kushwah
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B.Tech.
Gender: Male
DOB: 22/05/2002

Examination	University	Institute	Year	CPI/%
Matriculation	State Board (M.P.)	I.P.S School	2018	97.40%
Intermediate	State Board (M.P.)	Govt. School Khareih	2020	90.40%

Scholastic achievements

- Secured **98.53** percentile in JEE Advanced and **99.52** percentile in JEE Mains among **0.93+** million aspirants *(2021)*
- Secured the **top(1st)** rank in entire region (**1000+** students) in the state Boards of **12th**-grade examination *(2020)*

Professional Experience

Data Scientist (Platform Team) [*July'25-Present*]

L&T Finance Pvt. Ltd. | Streamlined cross-department automation workflows across the company.

- Worked on **24/7 automated logging** system sending pipeline error alerts with solutions, cutting **costs** by **50%**
- Developed a **chat-based Credit Risk & Collections Intelligence system** using **MCP** and A2A architectures, enabling **non-technical** users to query and analyse **10+** million rows data by simple natural-language conversation.
- Built an automated **email management system** using **Transformer embeddings** & **SVM** classification achieving **91%** accuracy, generating response, processing **1,500+** customer emails daily with automatic **assigning**
- Fine-tuned a **company-specific SLM** using **Q-LoRA** and **Reinforcement Learning** based instruction tuning for retrieval and better chatbot response, achieving **0.87 cosine similarity, 0.39 BLEU, and 0.51 ROUGE-I**

Internships and Projects Experience

AI (Finance Agentic Systems) & Data Intern (Vidur Research) **Part time* [*Sep'25-Dec'25*]

- Designed a **MAS**-based agentic retriever that queries **1M+ news** data**, fundamentals,** and **100+ web pages**
- Fine-tuned **DeepSeek**-VL-R1 model with **QLoRA** to precisely get numerical values from documents (**200+** words)
- Generated **Nifty50** investment reports in **<10** min with **8.5/10** GPT-rated data **hygiene** using a **fine**-tuned model.

Generative AI Internship (OXFIN TECHNOLOGIES Pvt. Ltd.) [*Jul'24-Sep'24*]

- Developed a project using **LLM** models to automatically **rewrite sentences**, adhering with over **40** instructions
- Experimented with the various **cost-efficient** models, including **LLaMA 3:8B**, **LLaMA 3:7B**, and **Mistral 7B**

Generative AI Projects (Freelance & Research) [*March'25–Nov'25*]

- **Freelance**-Built an agentic **AI orchestrator** integrating **25+** travel **APIs** (Flights, Hotels, Trains) using LLM-based task decomposition, implemented via LangGraph/LangChain and a **custom** architecture for speed and flexibility.
- Programmed a **prompt-based** system using fine-tuned **local** models to generate **React landing** pages in **5** min
- Built a silent-speech recovery model using **LipNet** + 3D **CNN-GRUs**, delivering fast and accurate **real-time TTS**
- Implemented **MAV** detection in **C++**, detecting **UAVs** at **89% accuracy** and **25 FPS** using motion & appearance

Computer Vision Internship (Pharma llama Pvt. Ltd.) [*Jun'24-Aug'24*]

- Manually created a dataset of **0.4 million** images with **labels** and **bounding boxes** to detect general medicines
- Used various models (Faster-RCNN, YOLOv8), got **Faster-RCNN** with the high accuracy (**95.5%+**) and **speed**

Research Projects

Revolutionizing Refueling: Automatic Fuel Filling Arm (Guide: Prof. Anirban Guha) [*Aug'24-Nov'24*]

- Developed techniques for detecting **fuel** tanks (**>85%** confidence) and determining their angles to guide the **robotic arm**
- Fine-tuned pretrained YOLOv8 for **100+** epochs, with **~0.6s** per-image inference time, cutting fuel filling time by **~30%**

High Accuracy Satellite Image classification (Guide: Prof. Biplab Banerjee) [*Apr'24-Aug'24*]

- Increased accuracy by **(1.5-2%)** using **Drop-Block** with various **(4+) CNN** architectures in **classifying satellite** images
- Achieved remarkable closeness (**97%**) across multiple datasets, including **P**avia **U**niversity, **H**ouston, **I**ndia **P**ines,
- Used methods like **E**fficientNet **B7**, **I**ntegrated **ViT**, **M**ultihead attention, enhanced **69.4%** for **Resnet-18** on CIFAR100

Junior Propulsion Engineer (Position of Responsibility- Hyperloop Tech Team) [*Aug'22-Feb'24*]

- Contributed to propulsion team for IIT Bombay's first **hyperloop pod**, focusing on speed (**500** km/h+) and efficiency

Technical Skills

Languages: Python, JavaScript, Java (Basics), C/C++ (Complete DSA), MATLAB, SQL, R, HTML, CSS, ReactJs

Libraries: Pytorch, TensorFlow, Keras, OpenCV, ScikitLearn, Seaborn, Numpy, Pandas, Nltk, Yolo, Roboflow

***ML courses:** Introduction of remote sensing and Image processing*, Introduction to Machine Learning*, Deep Learning - Theory and Practice*, Statistical Machine Learning and Data Mining*, 3 Specialization Courses in Machine Learning (Coursera), Embedded Control & Robotics, Numerical Analysis, Linear Algebra, Differential Equation, Computer Programming and Utilization, Quantum Physics

Extracurricular: Mentored **120+** dancers with **6**-member choreographer's team in Institute's **flagship** Dance show(2x)